```python
import cv2
import numpy as np
from matplotlib import pyplot as plt
left_img  = cv2.imread(r"C:\Users\sreev\Downloads\left.png")
right_img = cv2.imread(r"C:\Users\sreev\Downloads\right.png")
if left_img is None or right_img is None:
    print("□ Error: Could not load images!")
else:
    print(f"✓ Left image size:  {left_img.shape}")
    print(f"✓ Right image size: {right_img.shape}")
h = min(left_img.shape[0], right_img.shape[0])
w = min(left_img.shape[1], right_img.shape[1])
left_img  = left_img[:h, :w]
right_img = right_img[:h, :w]
left_gray  = cv2.cvtColor(left_img, cv2.COLOR_BGR2GRAY)
right_gray = cv2.cvtColor(right_img, cv2.COLOR_BGR2GRAY)
left_gray  = cv2.equalizeHist(left_gray)
right_gray = cv2.equalizeHist(right_gray)
print(f"✓ Working size: {left_gray.shape}")

stereo = cv2.StereoBM_create(numDisparities=128, blockSize=15)
stereo.setPreFilterType(1)
stereo.setPreFilterSize(5)
stereo.setPreFilterCap(61)
stereo.setTextureThreshold(507)
stereo.setUniquenessRatio(15)
stereo.setSpeckleRange(32)
stereo.setSpeckleWindowSize(100)
stereo.setMinDisparity(0)

disparity = stereo.compute(left_gray, right_gray)
print("✓ Disparity map computed!")

baseline     = 65.0
focal_length = 1000.0
disparity_float = np.float32(disparity) / 16.0
disparity_float[disparity_float <= 0] = 0.1
depth = (baseline * focal_length) / disparity_float
depth[depth > 50000] = 0
print("✓ Depth map calculated!")

depth_filtered = cv2.medianBlur(depth, 5)
# Remove noise
depth_filtered = cv2.bilateralFilter(depth_filtered, 9, 75, 75)
# Smooth edges
disparity_norm = cv2.normalize(disparity_float, None, 255, 0,
cv2.NORM_MINMAX, cv2.CV_8U)
depth_norm     = cv2.normalize(depth_filtered,  None, 255, 0,
cv2.NORM_MINMAX, cv2.CV_8U)
depth_colored  = cv2.applyColorMap(depth_norm, cv2.COLORMAP_JET)
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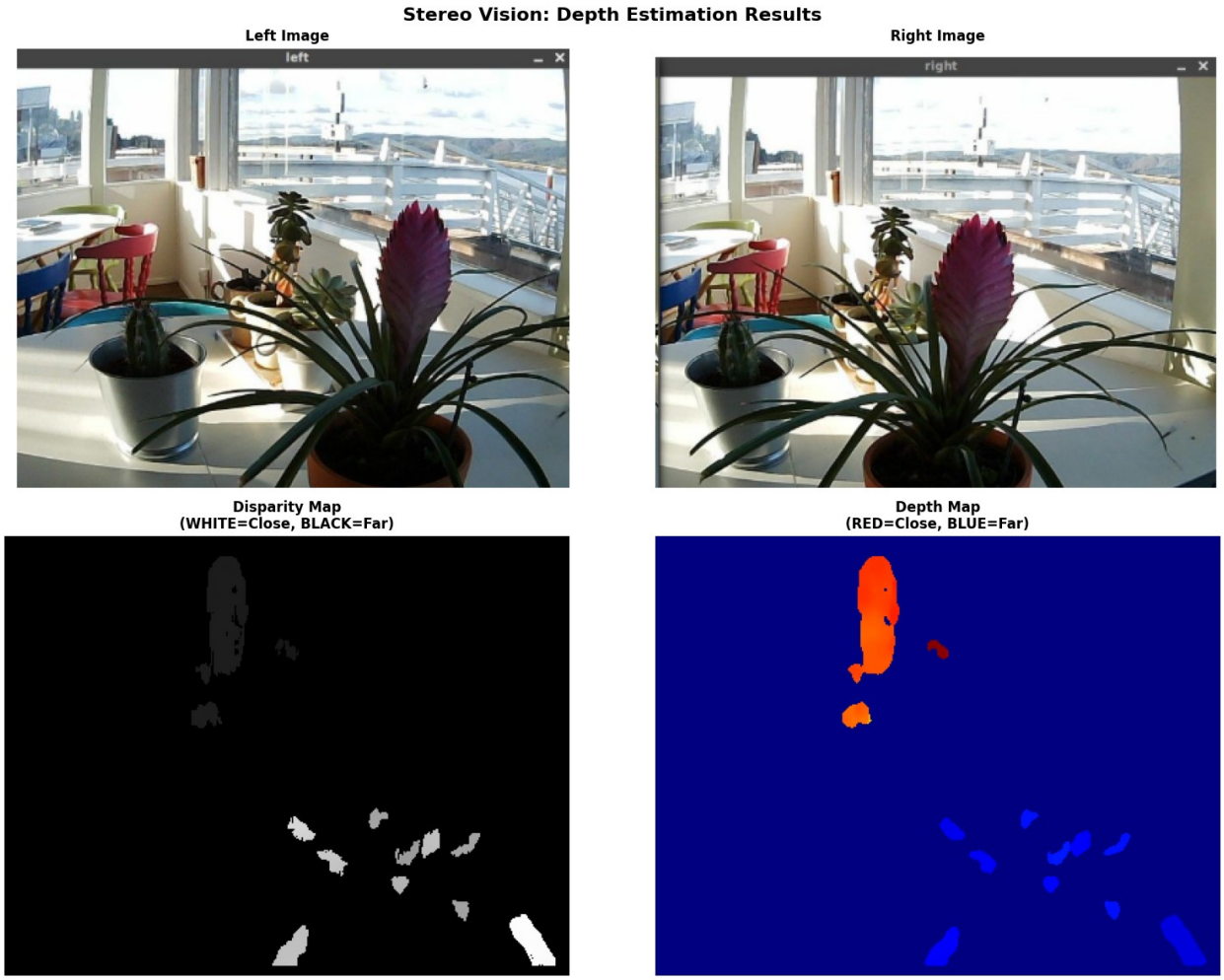
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle("Stereo Vision: Depth Estimation Results", fontsize=16,
fontweight='bold')
axes[0, 0].imshow(cv2.cvtColor(left_img, cv2.COLOR_BGR2RGB))
axes[0, 0].set_title("Left Image", fontsize=12, fontweight='bold')
axes[0, 0].axis('off')
axes[0, 1].imshow(cv2.cvtColor(right_img, cv2.COLOR_BGR2RGB))
axes[0, 1].set_title("Right Image", fontsize=12, fontweight='bold')
axes[0, 1].axis('off')
axes[1, 0].imshow(disparity_norm, cmap='gray')
axes[1, 0].set_title("Disparity Map\n(WHITE=Close, BLACK=Far)",
fontsize=12, fontweight='bold')
axes[1, 0].axis('off')
axes[1, 1].imshow(cv2.cvtColor(depth_colored, cv2.COLOR_BGR2RGB))
axes[1, 1].set_title("Depth Map\n(RED=Close, BLUE=Far)", fontsize=12,
fontweight='bold')
axes[1, 1].axis('off')
plt.tight_layout()
plt.show()

valid_depth = depth_filtered[depth_filtered > 0]
print("\n" + "="*50)
print("DEPTH STATISTICS")
print("="*50)
print(f"Min Depth  (Closest) : {np.min(valid_depth):.1f} mm")
print(f"Max Depth  (Farthest): {np.max(valid_depth):.1f} mm")
print(f"Mean Depth           : {np.mean(valid_depth):.1f} mm")
print(f"Median Depth         : {np.median(valid_depth):.1f} mm")
print(f"Valid Pixels         : {len(valid_depth)} /
{depth_filtered.size} "
      f"({len(valid_depth)/depth_filtered.size*100:.1f}%)")
print("="*50)

cv2.imwrite(r"C:\Users\sreev\Downloads\disparity_map.png",
disparity_norm)
cv2.imwrite(r"C:\Users\sreev\Downloads\depth_map.png",
depth_colored)
print("\n✓ Results saved to Downloads folder!")
print("  - disparity_map.png")
print("  - depth_map.png")
```

```
✓ Left image size:  (315, 408, 3)
✓ Right image size: (325, 405, 3)
✓ Working size: (315, 405)
✓ Disparity map computed!
✓ Depth map calculated!
```

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DEPTH STATISTICS
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Min Depth  (Closest) : 635.4 mm
Max Depth  (Farthest): 7019.5 mm
Mean Depth           : 3428.3 mm
Median Depth         : 5431.3 mm
Valid Pixels         : 4579 / 127575 (3.6%)
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✓ Results saved to Downloads folder!
  - disparity_map.png
  - depth_map.png
```